

AJAY VARADA

+1 404-203-2656 | ajayvrda@gmail.com | linkedin.com/in/ajay-varada | github.com/ajayvarada

EDUCATION

Georgia State University

Master of Science in Computer Science – CGPA: 4.0/4.0

Jul 2025 – Present

Atlanta, GA

Keshav Memorial Institute of Technology (KMIT)

Bachelor of Technology in Computer Science (Data Science) – CGPA: 3.8/4.0

Dec 2020 – Jul 2024

Telangana, India

TECHNICAL SKILLS

Languages: Python, Java, C++, C#, SQL, JavaScript

CS Fundamentals: Data Structures, Algorithms, OOP, Problem Solving, Debugging, Performance Optimization

Cloud & Tools: AWS, Git, GitHub, Azure DevOps (CI/CD), Linux/Unix, REST APIs

AI & ML: GenAI, LLM APIs, LangChain, PyTorch, TensorFlow, scikit-learn, NLP, Computer Vision

Databases: SQL (Query Optimization, Relational Design), NoSQL (MongoDB), Vector Databases, ETL Pipelines

Methodologies: Agile/Scrum, Code Reviews, TDD, Technical Documentation, Operational Excellence

WORK EXPERIENCE

Micron Technology

Jul 2024 – Jul 2025

Software Engineer I (Promoted from Intern)

- Developed and deployed a **Python-based NLP analytics tool** using **LLM APIs**, **Pandas**, and **scikit-learn** to automate analytical workflows at scale; adopted by **170+ engineers**, reducing per-task overhead by **40%** – debugged edge cases and iteratively improved reliability based on user feedback.
- Developed **automation workflows and ML model prototypes** in Python for network configuration and compliance validation; contributed to **Azure DevOps CI/CD pipelines** with Git version control, contributing to a **35%** reduction in release cycle time across **4+ internal services**.
- Debugged and optimized **CI/CD workflows** to improve pipeline stability, increasing system reliability by **30%**; participated in **50+ code reviews** with a focus on clean, maintainable code; recognized with **multiple Employee of the Month** awards.
- Implemented efficient **data processing pipelines** and optimized algorithms to handle **large-scale workloads** with low latency across distributed systems.

Micron Technology

Jan 2024 – Jul 2024

Software Engineering Intern

- Built **100+ automation scripts** integrated into **CI/CD pipelines** for enterprise network and security validation using **Python and SQL**; applied strong **problem-solving and debugging skills** to resolve complex edge cases – awarded **Best Intern of the Year** and converted to full-time **SDE I**.

DigiClinics Research and Services

Aug 2023 – Dec 2023

AI/ML Engineer Intern

- Built an end-to-end **ML pipeline** in **Python**, **PyTorch**, and **scikit-learn** achieving **~90% segmentation accuracy**; deployed a production system serving **2 hospitals** and **10+ institutions**, and co-authored a peer-reviewed manuscript (under review).

PROJECTS

AegisMem – LLM Agent Memory System | Python, LangChain, Flask, AWS, Vector DB, REST API 2025

- Designed and built a **scalable, distributed memory system** for long-running LLM agents using **vector databases** (FAISS/ChromaDB) and **semantic retrieval pipelines**; architected a **microservices-style Flask REST API** for memory ingestion and lifecycle management deployable on **AWS**.
- Implemented **graph-based and hash-indexed retrieval algorithms** for efficient memory lookup; applied **data structures** (priority queues, hash maps, trees) to optimize retrieval latency; open-sourced the codebase with structured documentation and authored a **research paper** on the architecture.

Ischemic Stroke Classification | Python, PyTorch, ResNet101V2, scikit-learn, SQL, Data Visualization 2022

- Trained ResNet101V2 on **30,000+ pathology image tiles** stored in a **SQL database** achieving **0.96 AUC** and **93% accuracy** (+20% over baseline); delivered visualizations communicating model performance to clinical stakeholders.
- Built a **modular, end-to-end ML pipeline** covering data ingestion, preprocessing, augmentation, model training, and evaluation using **PyTorch and scikit-learn**; designed for reproducibility and extensibility across additional pathology classification tasks.

Auto Image Annotations | Python, TensorFlow, SSD MobileNet, REST API, SQL 2022

- Developed an **object detection pipeline** (SSD MobileNet, 91 COCO classes, **0.68 mAP**) with a **SQL-backed data pipeline** and **REST API** inference layer; reduced annotation time by **90%**, saving 100+ engineering hours.